Mapping instruments used to assess the impact of patient education in chronic diseases: a scoping review

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Background

- PE is a key component of chronic disease management.
- Patient education (PE) aims to strengthen patients' self-management skills and confidence in managing their condition.
- However, it remains unclear whether existing generic instruments adequately capture the competencies targeted by PE programmes.

AIM: To identify generic instruments used to assess the impact of PE, map their outcome domains, and highlight measurement gaps.

Methods

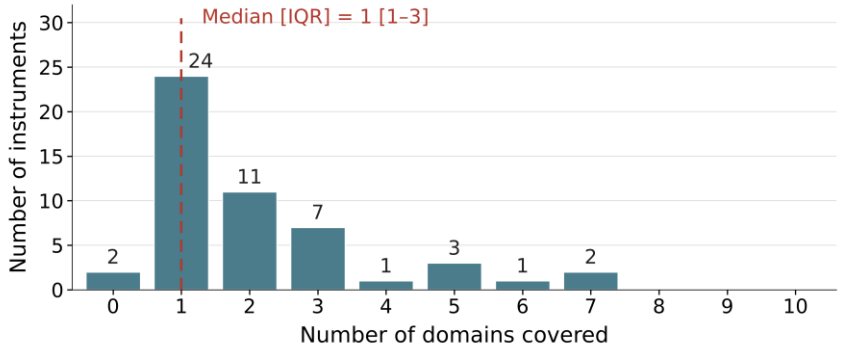
Scoping review followed JBI Guidelines[1]

- **Search** — Databases: PubMed, Embase, PsycINFO, and Cochrane; no language or date restrictions
- **Eligibility** — Studies using a generic instrument to assess PE in chronic disease populations
- **Selection** — AI-assisted screening (ASReview[2]) with human calibration
- **Analysis** — Instruments mapped to 10 PE competency domains (heuristic framework: WHO[3], Lorig & Holman[4]) using binary coding

Results

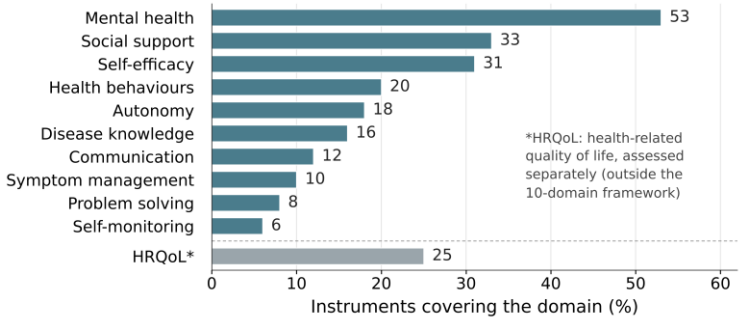
- 9,985 abstracts → 256 full texts → 77 studies from 26 countries
- 51 instruments | 13 condition groups
- **Most instruments cover few domains.** The median is 1 of 10 (IQR 1–3), and 86% of instruments (44/51) cover 3 domains or fewer.
- **No instrument covers all 10 domains.** The heiQ (Health Education Impact Questionnaire) and the PIH (Partners in Health scale), reach 7/10.

Fig 1: Distribution of instruments by number of domains covered (N=51)



- **Uneven coverage**: mental health most frequently assessed; self-monitoring, problem-solving, and symptom management least represented.

Fig 2 : Proportion of generic PE instruments assessing each competency domain(N=51)



Conclusion

- Findings suggest a potential misalignment between available generic PE instruments and WHO/Lorig self-management competency domains.
- Whether disease-specific instruments compensate for these gaps warrants further investigation.

Discussion

- Instrument design matters: the most comprehensive instruments (heiQ, PIH) were developed specially to evaluate self-management or PE.
- Generic instruments mainly captured psychological and social outcomes more than the skills PE programmes target
- **Limitations:** only generic instruments were included; binary coding indicates whether a domain is assessed, not how well; the heuristic framework was not empirically validated for instrument mapping.

References:
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3. World Health Organization Regional Office for Europe. *Therapeutic Patient Education: Continuing Education Programmes for Health Care Providers in the Field of Prevention of Chronic Diseases. Report of a WHO Working Group.* Copenhagen: WHO Regional Office for Europe; 1998.
4. Lorig KR, Holman H. Self-management education: history, definition, outcomes, and mechanisms. Ann Behav Med. 2003;26(1):1–7. doi:10.1207/S15324796ABM2601_01



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